Supplement Table 1. Characteristics of included studies

Gulani et al (2019) [30]	
Methods	Study design: Parallel-group randomised controlled trial (quasi-randomization) Study dates: January 2015–November 2016 Setting: single institution Country: India Study endpoint: 6 months
Participants	Inclusion criteria: patients who underwent ligual mucosa graft urethroplasty Exclusion criteria: patients with oral cancer, oral ulcer, submucosal fibrosis, poor mouth opening, previous oral surgery, and bilateral lingual harvest site Screened: not reported; Eligible: 42 Age (mean in years±SD): Non-closure 37.7 (SD: 9.3); Closure 36.5 (SD: 9.1) Stricture location: not reported Graft width (cm±SD): Non-closure 1.84 (SD: 0.3); Closure 1.89 (SD: 0.3) Graft location and laterality: ventrolateral side tongue; unilateral Pain control: no information
Interventions	Graft harvest: The segment to be harvested was marked on the ventrolateral surface and infiltrated submucosally with adrenaline (1:100,000). The graft edges are incised with a scalpel and a full-thickness mucosal graft is harvested using sharp scissors beginning at the anterior land mark of the graft. Tip of the tongue was spared. After graft harvesting, donor site bleed was controlled with bipolar cautery. Intervention (n=21): Non-closure Control (n=21): Closure; closed with interuppted 3–0 chromic catgut suture
Outcomes	Primary and secondary outcome were not mentioned • Pain How measured: visual analog pain score Time points measured: day 0, day 3, 1 week, 1 month, and 6 months Time points reported: day 0, day 3, 1 week, 1 month, and 6 months • Difficulty in tongue protrusion • Swelling • Numbness in graft harvest site • Speech impairment • Regular diet How measured: Non-validated questionnaire Time points measured: day 0, day 3, 1 week, 1 month, and 6 months Time points reported: day 0, day 3, 1 week, 1 month, and 6 months
Funding sources	None
Declarations of interest	None
Notes	Language of publication: English

Risk of bias table		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	High risk	Quote: "The method of randomization was every alternate patient patient (1:1 ratio)" Comment: quasi-randomization
Allocation concealment (selection bias)	High risk	Quote: "The method of randomization was every alternate patient patient (1:1 ratio)" Comment: quasi-randomization
Blinding of participants and personnel (performance bias)	High risk	Quote: not avaliable Comment: There is no mention of blinding wihtin the report. Given the mode of randomization, the allocation would have been apparent to the surgeon at the time of surgery. The bias is likely to have material as prior knowledge of a patient's allocation may affect operative technque and extent of hemostasis employed.
Blinding of outcome assessment (detection bias)	High risk	Quote: not avaliable Comment: Considering outcome measures were investigated after surgery, it is not possible to blind outcome assessor or participants
Incomplete outcome data (attrition bias)	Low risk	Quote: not avaliable Comment: All participants completed the study and are included in analysis

The World Journal of MEN's HEALTH

Selective reporting (reporting bias)	Unclear risk	No protocol
Other bias	Low risk	No other apparent sources of bias detected

Muruganandam et al (2009) [31]	
Methods	Study design: Parallel-group randomised controlled trial (quasi-randomization) Study dates: July 2003–July 2005 Setting: single institution Country: India Study endpoint: 6 months
Participants	Inclusion criteria: Patients undergoing buccal mucosa graft substitution urethroplasty Exclusion criteria: not reported Screened: not reported; Eligible: 50 Age (mean in years±SD): Non-closure 35.4 (SD 11.5); Closure 35.2 (SD 13.8) Stricture location: panurethra, bulbar, penobulb Graft size (cm^2±SD): Non-closure 14.9 (SD: 6.8); Closure 8.6 (SD: 3.2) Graft location and laterality: Cheek 44, lip 6, unilateral 34, bilateral 16 Pain control: No information
Interventions	Graft harvest: "Parotid duct opening identified and protected; Plane infiltrated with 1% xylocaine and adrenaline; Dissection off buccinator with fine scissor"; "After the graft is harvested, perfect hemostasis is achieved with bipolar electrocautery and an adrenalin soaked gauze for 4–6 hours." Intervention (n=25): Non-closure Control (n=25): Closure; closed with continuous interlocking sutures (3–0 vicryl)
Outcomes	Primary and secondary outcome were not mentioned • Pain How measured: 5 point non-validated scale Time points measured: POD 1–5, 3 months, 6 months Time points reported: POD 1, POD 2, POD 5, 6 months • Difficult mouth opening • Numbness How measured: Non-validated questionnaire (not reported) Time points measured: not reported Time points reported: POD 1, 1 week, 6 months • Salivary problems How measured: not reported Time points measured: POD 1, month 3, month 6 Time points reported: not reported • Regular diet (liquid/solid) How measured: not reported Time points measured: POD 1, month 3, month 6 Time points reported: POD 1 (liquid), POD 3 (solid)
Funding sources	None
Declarations of interest	None
Notes	Language of publication: English

Risk of bias table		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	High risk	Quote: "The method of randomization was every alternate patient patient (1:1 ratio)" Comment: quasi-randomization
Allocation concealment (selection bias)	High risk	Quote: "The method of randomization was every alternate patient patient (1:1 ratio)" Comment: quasi-randomization
Blinding of participants and personnel (performance bias)	High risk	Quote: not avaliable Comment: There is no mention of blinding wihtin the report. Given the mode of randomization, the allocation would have been apparent to the surgeon at the time of surgery. The bias is likely to have material as prior knowledge of a patient's allocation may affect operative technque and extent of hemostasis employed.

Blinding of outcome assessment (detection bias)	High risk	Quote: not avaliable Comment: Considering outcome measures were investigated after surgery, it is not possible to blind outcome assessor or participants
Incomplete outcome data (attrition bias)	Low risk	Quote: not avaliable Comment: All participants completed the study and are included in analysis
Selective reporting (reporting bias)	High risk	Quote: not avaliable Comment: Selective choice of reported data noted for numerous outcomes when comparing reported timing of measurements and reported outcomes: Pain data missing from month 3, Return to PO intake data not reported on POD 2, 4, 5 and months 3 and 6. In the absence of a protocol the extent of additional selective reporting is uncertain.
Other bias	Low risk	No other apparent sources of bias detected

Rourke et al (2012) [27]	
Methods	Study design: Parallel-group randomised controlled trial Study dates: June 2006–June 2008 Setting: single institution Country: Canada Study endpoint: 6 months
Participants	Inclusion criteria: Men undergoing urethroplasty, unilateral grafts Exclusion criteria: Bilateral graft, non-english speaking, cognitive impairment, history of oral cancer Screened: 56; Eligible: 50 Age (mean in years±SD): Non-closure 44.4 (SD 14.3); Closure 43.8 (SD 14.3) Stricture location: bulbar Graft size (cm^2±SD): Non-closure 15.04 (SD: not reported); Closure 15.25 (SD: not reported) Graft location and laterality: Cheek; unilateral Pain control: No information
Interventions	Graft harvest: The graft template was marked in a rectangular fashion on the inner cheek to a stanardized size of 6 cm long by 2.5 cm wide. Buccal infiltration with 1% lidocaine with epinephrine. Sharp dissection of graft. "Bipolar electrocautery was used for hemostasis." Intervention (n=24): Non-closure Control (n=26): Closure; closed with running 3–0 chromic suture
Outcomes	Primary outcome • Pain How measured: 10 point numeric pain scale Time points measured: POD 1–7, 1–6 months Time points reported: POD 1, POD 6, 6 months Secondary outcome • Interval to full mouth opening • Numbness • Salivation • Regular diet How measured: Non-validated questionnaire Time points measured: POD 1–7, 1–6 months Time points reported: POD 1, POD 6, 1 month (opening, regular diet), 6 months (numbness), salivation: not reported
Funding sources	Not reported
Declarations of interest	Not reported
Notes	Language of publication: English

Risk of bias table		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Quote: "Randomization was performed on day of surgery using a standard block randomization technique" Comment: Authors provide inadequate detail to determine whether an appropriate random sequence generation process was utilized

Allocation concealment (selection bias)	Low risk	Quote: "The patients were offered enrollment into the study by the surgeon at the intial consultation…Randomization was performed on day of surgery using a standard block randomization technique" Comment: Since enrollment occurred prior to randomization, the risk of selection bias is generally low however given the use of block randomization there is some risk that allocation could be predicted based on knowledge of prior interventions.
Blinding of participants and personnel (performance bias)	High risk	Quote: "An unmarked sealed envelope with the group assignment was brought to the operating suite and opened immediately before surgery by the attending physician" Comment: Personel is unblinded prior to completion of the graft harvest and achievement of hemostasis. Knowledge of patient allocation prior to completion of both harvest and hemostasis introduces the possibility of systematic differences between the two groups that are likely to have affected outcomes of interest (*e.g.*, less extensive graft resection or more aggressive hemostasis in non-closure group would reasonably affect postoperative pain)
Blinding of outcome assessment (detection bias)	High risk	Quote: not avaliable Comment: Considering outcome measures were investigated after surgery, it is not possible to blind outcome assessor or participants
Incomplete outcome data (attrition bias)	Low risk	Quote: not avaliable Comment: All participants completed the study and are included in analysis
Selective reporting (reporting bias)	High risk	Quote: not avaliable Comment: Selective reporting of outcome data noted for numerous outcomes when comparing reported timing of measurements and timing of reported outcomes (Pain measured on POD 1–7 and months 1–6 but outcomes reported only on POD 1, POD 6 and 6 months; Mouth opening measured on POD 1–7 and months 1–6 but outcomes reported only on POD 1, PO6 and 1 month; Numbness measured on POD 1–7 and months 1–6 but outcomes reported only on POD 1, POD 6 and 6 months; Salivation measured on POD 1–7 and months 1–6 but outcomes not reported; PO intake measured on POD 1–7 and months 1–6 but outcomes reported only on POD 1–7 and month 1)
Other bias	Low risk	No other apparent sources of bias detected

Soave et al (2018) [29]	
Methods	Study design: Parallel-group randomised controlled trial Study dates: October 2014–December 2015 Setting: single institution Country: Germany Study endpoint: 6 months
Participants	Inclusion criteria: Male patients >18 year of age with urethral stricture disease Exclusion criteria: Any previous treatment with BMGU, known or suspected concomitant oral diseases (gingivitis and caries), psychiatric disorders or cognitive impairment, chronic pain, bilateral buccal mucosa graft harvest, and non–German-speaking patients. Screened: not reported; Eligible: 163 Age (median): Non-closure 53; Closure 55 Stricture location: bulbar, penile Graft size (cm^2±SD): Non-closure 6.75 (SD: not reported); Closure 7.5 (SD: not reported) Graft location and laterality: Cheek; unilateral Pain control: "In brief, patients received analgesics according to the WHO Analgesic Ladder consisting of non-steroidal anti-inflammatory drugs (NSAIDs; *i.e.*, novalgin) and paracetamol, combined with weak opioids (*i.e.*, tramadol) according to the needs of patients. According to institutional standards, all patients performed daily oral rinsing with chamomile and cooling of the cheek from postoperative days 1–5."
Interventions	Graft harvest: Buccal mucosa infiltrated with 2% lidocaine and adrenaline and harvested in ovoid shape from inner cheek - width 15 mm, "bipolar electrocautery was used for hemostasis", "one piece of cottonoid gauze placed at harvest site in both groups and removed at the end of the case" Intervention (n=82): Non-closure Control (n=81): Closure; closed with interrupted 4–0 monofil sutures

Outcomes	Primary outcome • Intensity and quality of pain How measured: Intensity: unidimensional single 11-point NRS; Quality of pain: multidimensional Shoft-Form mcGill Pain Questionnaire (SF-MPQ) consisting of a sneosry and an affective pain subscale with 11 and 4 items Time points measured: POD 1, 5, 21, 3 months, 6 months Time points reported: POD 1, 5, 21, 3 months, 6 months Secondary outcome • Impairment of mouth opening; Numbness; Salivation; Regular diet; Oral bleeding; Impairment of smiling; Impairment of whistling; Oral Swelling; Slurred speech; Burden in daily life How measured: Questionnarie - 5 stage scale (0=not at all, 1=a bit, 2=moderately, 3=much, 4=very much Time points measured: POD 1, 5, 21, 3 months, 6 months Time points reported: POD 1, 5, 21, 3 months, 6 months
Funding sources	None
Declarations of interest	None
Notes	Language of publication: English

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "randomized to the Closure group or Non-closure group in a 1:1 ratio using a computer-based randomization list containing consecutive numbers" Comment: Approrpriate use of a computer random number generator
Allocation concealment (selection bias)	Low risk	Quote: "When eligibility criteria were met, physicians of the special consultation hour for reconstructive urology of our department enrolled patients in the trial. Afterward, participating patients were randomized." Comment: 1:1 randomization took place after enrollment in a manner that neither participants nor investigators enrolling participants could forsee assigment prior to enrollment
Blinding of participants and personnel (performance bias)	High risk	Quote: "The surgical nursing staff kept the randomization list with the specific coding for Closure versus Non-closure in a locked drawer in a separate nursing-staff room in the OP wing. Only during the buccal mucosa harvest, the surgical nursing staff revealed whether the number encoded for Closure or Non-closure of the donor site." Comment: Personel is unblinded prior to completion of the graft harvest and achievement of hemostasis. Knowledge of patient allocation prior to completion of both harvest and hemostasis introduces the possibility of systematic differences between the two groups that are likely to have affected outcomes of interest (*e.g.*, less extensive graft resection or more aggressive hemostasis in non-closure group would reasonably affect postoperative pain)
Blinding of outcome assessment (detection bias)	High risk	Quote: not available Comment: Considering outcome measures were investigated after surgery, it is not possible to blind outcome assessor or participants
Incomplete outcome data (attrition bias)	High risk	Comment: About 38% of patients were missing at 6 months follow up.
Selective reporting (reporting bias)	Unclear risk	No protocol
Other bias	Low risk	No other apparent sources of bias detected

Wong et al (2014) [28]

Methods	Study design: Parallel-group randomised controlled trial Study dates: 2008–2013 Setting: single institution Country: United Kingdom Study endpoint: 12 months

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Participants	Inclusion criteria: Male had a single stricture of any aetiology amenable to buccal mucosal augmentation, stricture length no greater than 5 cm Exclusion criteria: Prior treatment with buccal graft, bilateral cheek, lingual or lip harvests Screened: 70; Eligible: 41 Age (median): Non-closure 41; Closure 47 Stricture location: No information Graft size (cm^2±SD): Non-closure 8.8±1.5; Closure 7.8±1.5 Graft location and laterality: Cheek; unilateral Pain control: No information
Interventions	Graft harvest: "Stenson's duct was marked and the graft was outlined. All grafts were 1.5x5 cm. The submucosa infiltrated with bupivicaine: Adrenaline 1:2,000,000 solution. The graft margin was sharply incised and bipolar diathermy utilized to cauterize the bleeding sites. If the patient was randoized to primary closure, it was achieved with 4–0 vicryl rapide suture in a continuous stitch. In patients in the open arm the defect was soaked in an adrenaline soaked gauze for the remainder of the procedure and removed at the end" Intervention (n=20): Non-closure Control (n=21): Closure
Outcomes	Primary outcome • Pain; Numbness; Mouth tightness; Regular diet How measured: 10 point numeric pain scale Time points measured: POD 1, 3, 3 weeks, 3 months, 6 months, 9 months, 12 months Time points reported: POD 1, 3, 3 weeks, 3 months, 6 months, 9 months, 12 months
Funding sources	Not reported
Declarations of interest	None
Notes	Language of publication: English

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote: "Once the graft had been harvested, the randomization process was carried out, which involves a member of staff selecting an unmarked envelope containing either 'closure' or 'open' in a 1:1 ratio" Comment: Unclear if a proper random sequence generation took place as there is no mention of a shuffling process
Allocation concealment (selection bias)	Unclear risk	Quote: "Once the graft had been harvested, the randomization process was carried out, which involves a member of staff selecting an unmarked envelope containing either 'closure' or 'open' in a 1:1 ratio" Comment: No mention whether the envelopes were sequentially numbered, opaque and sealed
Blinding of participants and personnel (performance bias)	Low risk	Quote: Surgeon unblinded "once the graft had been harvested" and "hemostasis was carried out using bipolar diathermy, if necessary" Comment: Considering the timing of randomization and allocation only after harvest and hemostasis and been performed, the risk of performance bias is low
Blinding of outcome assessment (detection bias)	High risk	Quote: not available Comment: Considering outcome measures were investigated after surgery, it is not possible to blind outcome assessor or participants
Incomplete outcome data (attrition bias)	High risk	7 patients lost to follow up (5 [24%] closure, 2 [10%] non-closure)
Selective reporting (reporting bias)	Unclear risk	No protocol
Other bias	Low risk	No other apparent sources of bias detected

POD: postoperative day, SD: standard deviation.